
AUDIO AMPLIFIER DESIGN AND SIMULATION

This assignment is designed to meet ABET performance standards for ECE with the following outcomes / performance indicators:

- Meets ECE Outcome #1: An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics. Performance indicators:
 - 1.2 Solves complex engineering problems.
 - 1.3 Applies principles of engineering to the solution of an engineering problem.
 - 1.4 Applies principles of science and mathematics to the solution of an engineering problem.
- Meets ECE Outcome #2: An ability to apply the engineering design process to produce solutions that meet specified needs with consideration for public health and safety and global, cultural, social, environmental, economic, and other factors as appropriate to the discipline. Performance indicators:
 - 2.1 Uses an engineering design process to develop a solution according to problem constraints.
 - 2.2 Applies the engineering design process to produce solutions that meet specified needs.
- Meets ECE Outcome #3: An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions. Performance indicators:
 - 3.1 Develops and conducts appropriate experimentation.
 - 3.3 Analyzes and interprets data.
 - 4.4 Uses engineering judgment to draw conclusions.

PROJECT

Your task is to develop an audio amplifier product. The amplifier will use the LM386 Low Voltage Audio Power Amplifier from Texas Instruments. The datasheet for this part is included in the Appendices. You will use the “LM386 Gain = 20” amplifier topology application described in the datasheet as a starting point. You will include a bypass capacitor from pin 7 to ground, and you will have different component values available than those shown in the datasheet. Based on this basic design, you will perform the following design and simulation tasks.

A. DESIGN AND SIMULATION

You are to use the application design in the specification sheet below to design and simulate the audio amplifier using LTSPICE. We will review basic amplifier terminology in the ECE380 lab session. Tutorials on how to perform the SPICE simulations will be available on Canvas. You will be required to simulate the design to determine the amplifiers’ sensitivity to various component choices, implement your choices into the design, and characterize the final design’s performance. The project Requirements Specifications are provided in Table 1.

TABLE 1. AUDIO AMPLIFIER DESIGN SPECIFICATIONS

Marketing Requirements	Engineering Design Requirements	Justification
1,2	1. Total harmonic distortion (THD) < 1%, up to the 5th harmonic, 1kHz input, 8 Ω load, at 1 W output power (with 12V supply).	The human ear can easily detect distortion greater than 1% 1W can produce an adequate volume to fill a small room.
1	2. Input-referred noise < 75 nV/ $\sqrt{\text{Hz}}$, 100 Hz to 20 kHz	The human ear will easily detect noise if the signal to noise ratio is less than 40dB
1	3. Power supply rejection ratio (PSRR) > 60 dB, 100 Hz to 20 kHz	
1	4. Common mode rejection ratio (CMRR) > 60 dB, 100 Hz to 20 kHz	
1	5. The gain should be constant within $\pm 3\text{dB}$ over the range 100Hz to 20kHz, with an 8 Ω load	This will provide consistency over the range of human hearing
3,4	7. Maximum circuit board size 1.25 by 2.0 inches (This will be part of the PCB layout assignment)	Will fit within a typical portable speaker enclosure
4	8. The manufacturing cost should not exceed \$6 (based on build quantity of 1000) You do not need to consider this currently.	This will ensure that the product is competitive with other products of the same class
5	9. Efficiency > 40% at Pmax power output (the power at which THD=1%), 1kHz input, 8 Ω load with 12V supply	Other amplifiers of this class can achieve this. It is significantly lower than the theoretical maximum efficiency.
6	10. The max voltage gain should be at least 26dB	Most portable devices have signal levels above 500mV
6	12. The AC input impedance should be > 1k Ω	Compatibility with headphone or line-out sources
5,6	13. The AC output impedance should be < 1 Ω	For compatibility with speakers (also improves efficiency)
6	14. The amplifier should <u>function</u> (but will not meet the 1W power specification) with supply voltages between 6V and 12V	Typical DC power supply voltages
Marketing Requirements 1. The amplifier should have good sound quality 2. The amplifier should have adequate power to be heard across an average size office or room 3. The amplifier should be small 4. The amplifier should have a low cost 5. The amplifier should have moderate efficiency 6. The amplifier should be compatible with most audio devices		
Customer Constraints 1. The amplifier should be based on the LM386 integrated circuit		

2. The amplifier should use components available in the senior design lab.
--

Using LTSPICE, you will perform a Sensitivity Analysis of the device and generate a table of parameters and performance as shown in Table 2. A **sensitivity analysis** should describe how circuit component values affect the critical performance parameters of the amplifier. Use a table like Table 2 to summarize the significant relationships that you discovered. Place an up arrow $\uparrow$ in the box to indicate a proportional dependency (increasing the component value increases the value of the performance parameter) and a down arrow $\downarrow$ to indicate an inverse dependency. You may use other symbols to describe other types of dependencies that you observe but be sure to clearly explain what each symbol represents. You may also want to differentiate strong from weak dependencies by using multiple arrows to indicate the relative strength. Table 3 gives the components and the ranges over which they should be evaluated.

TABLE 2: SENSITIVITY MATRIX EXAMPLE

Parameter	Cbyp	Cb	Vs	VsigDC	RLoad	Rpot
Low cutoff frequency, f_L						X
High cutoff frequency, f_H						X
CMRR						X
PSRR						X
Input Referred Noise. (IRN)						
Maximum Output power @ 1kHz and 1% THD						X

TABLE 3: COMPONENT DESCRIPTIONS

Name	Description	Range
Cbyp	Amplifier Bypass capacitor	10uF max
Cb	Output DC blocking capacitor	220uF max
Vs	Magnitude of the power supply	6V to 12V
VsigDC	DC offset (Common Mode) value of the input audio source	-200mV to 200mV
RLoad	The load resistance (speaker)	4 Ω , 8 Ω , 16 Ω
Rpot	The potentiometer resistance	2k Ω to 50k Ω

SIMULATION GUIDELINES

You will need to run multiple types of simulations to evaluate the following metrics:

1. Frequency response (100 Hz to 100 kHz) - you may want to sweep a wider frequency for the sensitivity analysis.
2. Common mode rejection ratio (CMRR) over a range of 100 Hz to 100 kHz
3. Power supply rejection ratio (PSRR) over a range of 100 Hz to 100 kHz
4. Input-referred noise.
 - a. Measured with Vol parameter set to 0.5, (50% volume).
 - b. Rpot range $2k\Omega$ to $50k\Omega$
 - c. Characterized in terms of $nV/\sqrt{Hz}$
5. Output power.
 - a. Use average (or real) power, *average*($V \cdot I$), for all power calculations.
 - b. Output power will be defined as the maximum real power consumed by the load while keeping the harmonic distortion at 1% or less (see item 6 below).
 - c. Use a 1kHz sinewave as the input signal VsigAC. Set the volume control to maximum and adjust VsigAC to vary the output power. This analysis will require that you use two .step directives, one for VsigAC and another for the parameter that you are exploring. It will generate a lot of simulation data. You should use a software tool (Excel, MATLAB, etc. ...) to process this data or you will get overwhelmed.
6. Harmonic distortion, you will need to measure Total Harmonic Distortion (THD) to characterize the maximum power, Pmax.
 - a. Compute harmonic distortion using the .four spice directive specifying 5 harmonics, 1kHz-5Khz. The LTspice command “.four 1k 5 V(out) “calculates the THD at the output node using the following equation:

$$\%THD = \frac{\sqrt{V_{rms2}^2 + V_{rms3}^2 + V_{rms4}^2 + V_{rms5}^2}}{V_{rms1}} \cdot 100$$

Where V_{rms1} is the RMS amplitude of the fundamental (1kHz), and V_{rmsn} is the RMS amplitude of the nth harmonic (2kHz-5Khz). It is a measure of the output power contained in the undesirable harmonics relative to the desired fundamental power.

Items 1-3 should be evaluated using AC analysis (See the Basic Simulation Video and CMRR_PSRR Video).
Item 4 uses a special form of AC analysis (the .noise command) (See the NoiseSimulation Video).
Items 5 and 6 require a transient analysis (See the Advanced Simulation Video).

FINAL DESIGN VERIFICATION OF PERFORMANCE

1. When you have completed the sensitivity analysis and chosen your final component values you will need to verify that your design meets all the specifications in Table 1. Most of these specifications can be verified using the same simulations as the sensitivity analysis without having to vary the component values – these are fixed to the values you choose based on the sensitivity analysis, etc.
2. Specification 9, efficiency, can be verified by measuring (using a measure statement in LTspice) the power dissipated by the 8Ω load divided by the average power delivered by the 12V power source.
3. Specification 12, AC input impedance, can be measured using $R_{in}=V_{in}/I_{in}$ at 1kHz with an AC analysis.
4. Specification 13, AC output impedance, can be measured by changing the RLoad value (i.e., from 4Ω to 16Ω) and measuring the change in the load current and voltage: $R_{out} = \frac{\Delta V_{load}}{\Delta I_{load}}$ at 1kHz AC.
5. In addition to verifying the specifications you should create a plot that shows the change in THD as the output power ranges from .1W to $1.1 \cdot P_{max}$ for an 8Ω load. P_{max} is defined as the maximum output power that your amplifier can supply at 1% THD.